First They Came for the Others: A Theory of Divide-and-Conquer

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November 2026

Motivation

"First they came for the socialists, and I did not speak out—because I was not a socialist... Then they came for me—and there was no one left to speak for me."

— Martin Niemöller

Motivation

- **Aggressors often "divide-and-conquer"**: consolidate power sequentially rather than at once. Potential victims fail to form a united front.
  - **Munich Agreement**: Sudetenland → concessions → broader threat.
  - **Crimea vs. Ukraine**: Local separatism, historical exceptionalism in 2014 → Minsk Agreements → full-scale invasion in 2022.
  - **"Salami Tactics"** (Rákosi, 1952): Authoritarians isolate and remove opposition groups slice by slice.
- **Why? Existing explanations**: Conflicting interests, free-riding, or mechanical coordination failures.
- **We focus on Inferential Friction**: Uncertainty about the aggressor's true intent. When the first victim is attacked, bystander must infer whether it reflects a **localized grievance** or a **systemic campaign**.

Overview

A Bayesian wedge: *"perhaps they came for them, not for me."*

1. **Model (Pure & Mixed Equilibria)**:
2. **Comparative Statics**:
   - Attack cost (c) $\uparrow$.
   - Correlation (ρ) $\downarrow$ and first-target specialness (a) $\uparrow$.
3. **Richer Environments**:
   - Behavioral responses (wishful vs. magical thinking).
   - Propaganda/Counter-propaganda, Treaties.
   - Downstream protection (more than 2 victims).

Literature

Model

- **Players**: Attacker A, potential victims V_1 and V_2.
- A**'s cost of attack**: $c \in (0,1)$ per victim. Material, diplomatic, reputational, and moral costs.
- A**'s private benefits**: (b_1, b_2) drawn from support

$$B := \{(b_1, b_2) : 0 \leq b_2 \leq b_1 \leq 1\}.$$

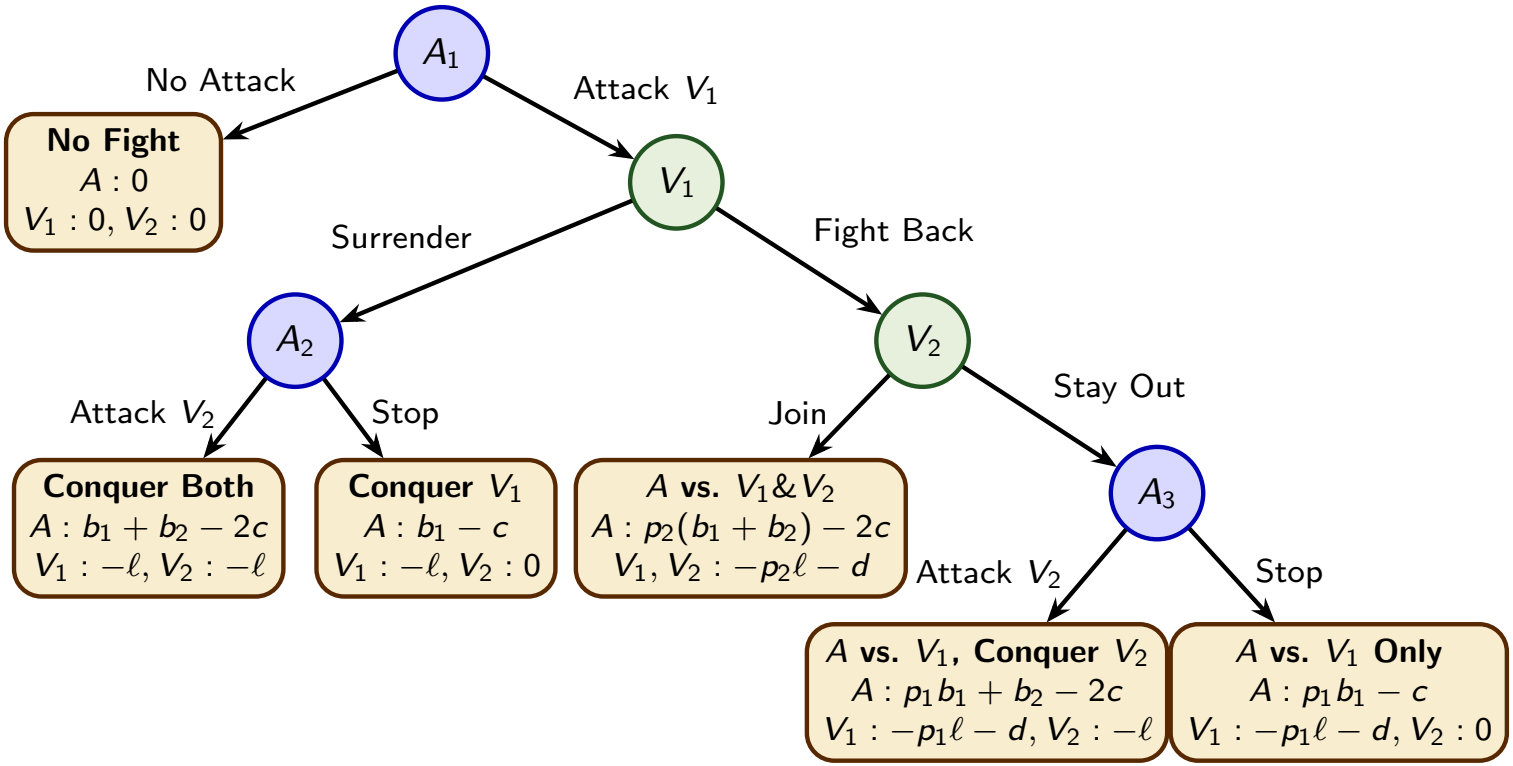
  - V_1 is the commonly recognized higher-stakes/more natural first target.
  - **Ordered i.i.d. Draws**: $b_1 = \max\{X_1, X_2\}$, $b_2 = \min\{X_1, X_2\}$ with $X_1, X_2 \overset{\text{i.i.d.}}{\sim} F$ on $[0,1]$ (e.g., uniform on B).

Model

- **For each victim V_i:**
  - If surrenders when attacked, loses $\ell > 0$ for sure.
  - If fights alone (at cost d), loses ℓ w.p. p_1; if fights with an ally, loses ℓ w.p. p_2, where $0 < p_2 < p_1 < 1$.
- **Assumption 1 (Resistance is effective only if coordinated)**

$$\underbrace{p_2\ell + d}_{\text{Joint fight}} < \underbrace{\ell}_{\text{Surrender}} < \underbrace{p_1\ell + d}_{\text{Lone fight}}.$$

Model



- **Sequential defense**: No simultaneous-move coordination failure.
- **Attack order**: Attacking V_1 first dominates simultaneous attack or attacking V_2 first (reveals $b_1 \geq b_2 \geq c$ and triggers joint resistance).

Model

- **Regular PBE**
  - "No signaling what you don't know": Since V_1 does not observe (b_1, b_2), an unexpected off-path fight by V_1 does not further alter V_2's belief beyond A's attack:
    $\mu_2(\cdot \mid A \text{ attacks } V_1, V_1 \text{ fights}) = \mu_2(\cdot \mid A \text{ attacks } V_1)$.
- **Full-Information Case**
  - If (b_1, b_2) is known, whenever $b_2 \geq c$, V_2 joins and V_1 fights. **Sequential conquest via passivity is impossible under full info.**

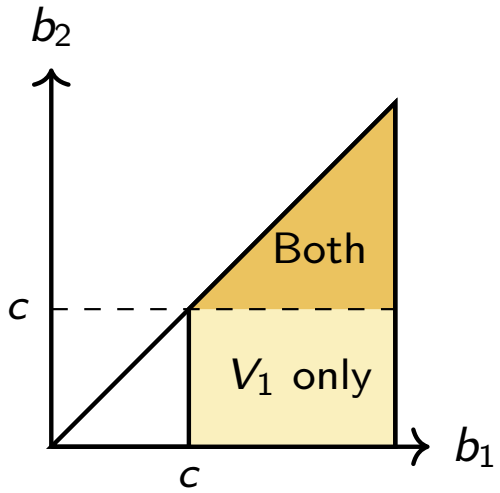
Pure-Strategy Equilibria

Define V_2's **posterior threshold** κ s.t. $\kappa \cdot \ell = p_2 \ell + d$.

1. Divide-and-Conquer (D&C)

- V_2 stays out $\Rightarrow$ V_1 surrenders.
- A attacks V_1 iff $b_1 \geq c$; later attacks V_2 iff $b_2 \geq c$.
- V_2's posterior risk of being next:

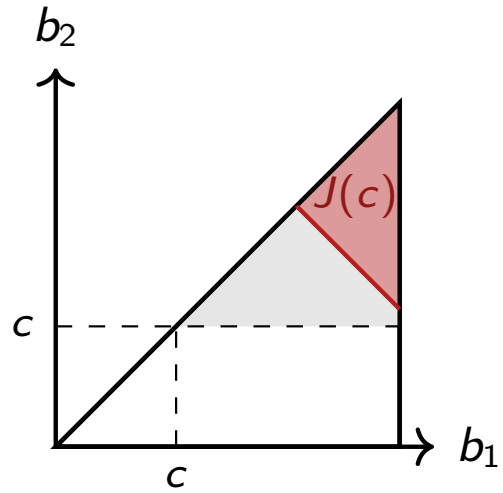
$$\phi(c) := \Pr(b_2 \geq c \mid b_1 \geq c) \leq \kappa.$$



2. Collective Resistance (CR)

- V_2 joins $\Rightarrow$ V_1 fights when attacked.
- A attacks iff two-front war is profitable: $J(c) := \{p_2(b_1 + b_2) \geq 2c\}$.
- V_2's posterior risk on $J(c)$:

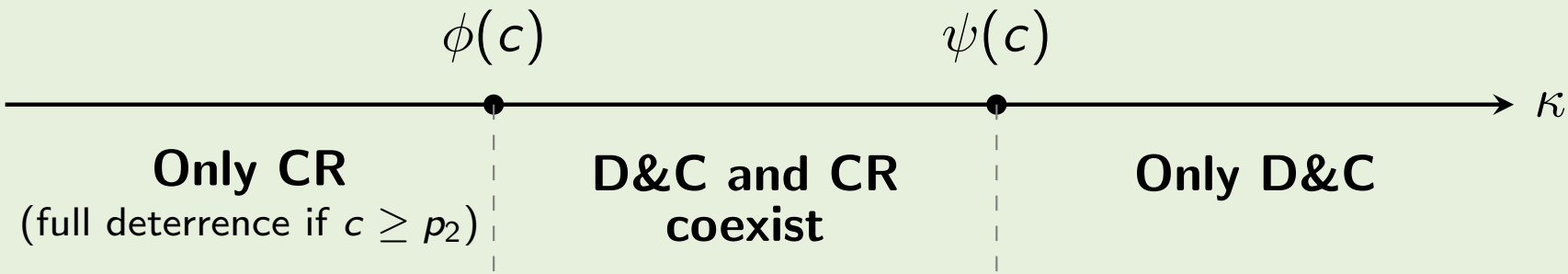
$$\psi(c) := \Pr(b_2 \geq c \mid J(c)) \geq \kappa.$$



Equilibrium Classification

Assumption 2: $\phi(c) \leq \psi(c)$. Implied by ordered i.i.d. draws.

Theorem 1 (Pure-Strategy Equilibrium Classification)



Mixed Resistance

Proposition 1 (Uncertain Collective Resistance)

When $\phi(c) < \kappa < \psi(c)$, $\exists$ a **mixed equilibrium** (V_1 fights w.p. α, V_2 joins w.p. β, resulting initial attack set $B_{\alpha,\beta}(c)$) s.t.

$$\Pr(b_2 \geq c \mid B_{\alpha,\beta}(c)) = \kappa, \qquad J(c) \subsetneq B_{\alpha,\beta}(c) \subsetneq \{b_1 \geq c\}.$$

- **Historical insight**: An aggressor hoping for D&C may encounter collective resistance (e.g., Russia's invasion of Ukraine).
- **Equilibrium Selection**:
  - Hereafter, we focus on the **attacker-preferred pure equilibrium** which selects D&C whenever available ($\phi(c) \leq \kappa$) and CR otherwise.

Mixed Equilibrium Details

Comparative Static I: Attack Cost c

What if the cost of attack $c \uparrow$? For ordered i.i.d. draws:

$$\phi(c) = \frac{1 - F(c)}{1 + F(c)}, \qquad \phi'(c) = \frac{-2F'(c)}{[1 + F(c)]^2} < 0.$$

A costly first attack looks less systemic:

- Ex ante, a higher cost c mechanically discourages aggression.
- **Conditional on observing an attack**, however, a higher cost leads V_2 to infer that A must have an exceptionally high target-specific stake in V_1 rather than a systemic campaign!
- When aggression is viewed as "unthinkable" or norm-breaking (e.g., 1936 Rhineland remilitarization, 2014 Crimea annexation), a first strike is more easily rationalized as a localized exception.

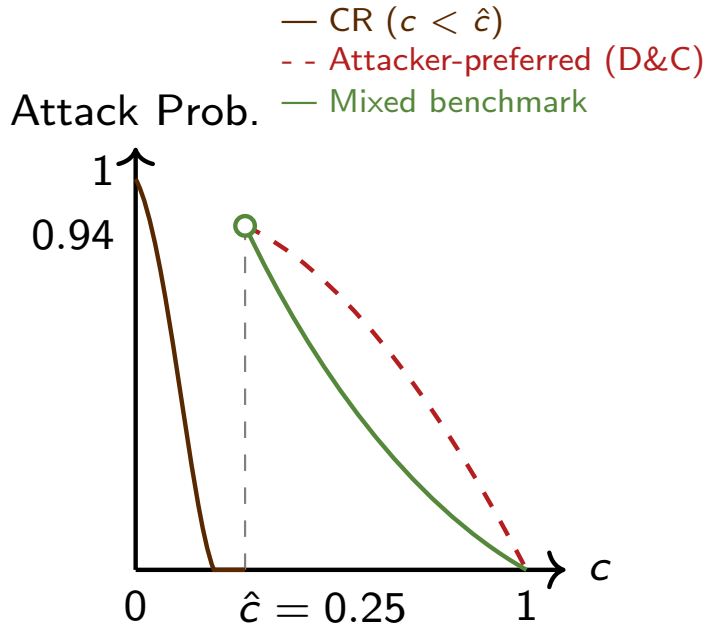
Comparative Static I: Upward Jump in Attacks at $\hat{c}$

Let $\hat{c}$ be the unique threshold where $\phi(\hat{c}) = \kappa$.

Proposition 2

1. **Below** $\hat{c}$: Only CR exists. Attack probability $\mu(J(c))$ decreases in c (reaches 0 at $c = p_2$ if $p_2 < \hat{c}$).
2. **At** $c = \hat{c}$: Attack probability jumps upward from $\mu(J(\hat{c}))$ to $1 - F_1(\hat{c})$!
3. **Above** $\hat{c}$: Attack probability decreases in c under both benchmarks.

Exposure of V_2 also jumps upward at $\hat{c}$ (from 0 to 0.56 in this example)!



Ordered i.i.d. uniform benefits,

$p_1 = 0.8, p_2 = 0.2, d/\ell = 0.4, \kappa = 0.6$

Comparative Statics II: Correlation

Proposition 3 (Correlation ρ between Victims' Fates)

Let primitive r.v.s (X_1, X_2) for ordered stats. have marginal F and copula $C_\rho(u, v) =: Pr(X_1 \leq F^{-1}(u), X_2 \leq F^{-1}(v))$.
Let $b_1 = \max\{X_1, X_2\}$ and $b_2 = \min\{X_1, X_2\}$:

$$\phi_\rho(c) = \frac{1 - 2F(c) + C_\rho(F(c), F(c))}{1 - C_\rho(F(c), F(c))}.$$

Suppose $C_\rho(u, u)$ is increasing in ρ for every $u \in (0, 1)$.

- **Higher ρ weakly raises $\phi_\rho(c)$ and shrinks the D&C region.**

Comparative Statics III: Specialness

Proposition 4 (Specialness a of V_1)

Let $a \geq 0$ be an additional **publicly known** benefit from conquering V_1 (first attack if $b_1 + a \geq c$):

$$\phi_a(c) = \frac{\Pr(b_2 \geq c)}{\Pr(b_1 \geq c - a)}.$$

- **Higher a weakly lowers $\phi_a(c)$ and expands the D&C region.**

Behavioral Response: Wishful Thinking

Belief distortion—systemic interpretation is too painful to accept.

Proposition 5 (Wishful Thinking) on attacker-preferred D&C.

- **Beliefs as consumption goods** (Brunnermeier & Parker, 2005).
- V_2 chooses subjective belief $m \in [\zeta\phi(c), \phi(c)]$ ($\zeta < 1$) to minimize total loss given decision $r \in \{S, J\}$:

$$L(r; \phi(c)) + \theta L(r; m), \quad \text{where } L(S; q) = q\ell \text{ and } L(J; q) = \kappa\ell.$$

- Staying out is optimal iff:

$$\phi(c) \leq \frac{1+\theta}{1+\theta\zeta}\kappa > \kappa \implies \hat{c}^{WT} < \hat{c}.$$

Expands the D&C region: Counter-propaganda must target comforting beliefs (i.e., tighten discipline ζ).

Behavioral Response: Magical Thinking

Misperceived consequence of inaction—hoping inaction buys safety.

Proposition 6 (Magical Thinking) on attacker-preferred D&C.

- **Misperceiving the causal effect of her own neutrality**.
- V_2 imagines staying out earns a "credit" $\omega > 0$, so A attacks V_2 later only if $b_2 \geq c + \omega$.
- Staying out is optimal iff the perceived risk under neutrality satisfies:

$$\Phi_\omega(c) := \Pr(b_2 \geq c + \omega \mid b_1 \geq c) \leq \kappa \implies \hat{c}_\omega^{MT} < \hat{c}.$$

Also expands the D&C region: Counter-propaganda must target the illusion that silence buys peace.

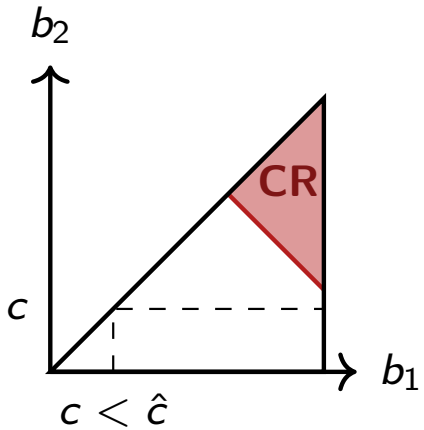
Propaganda and Counter-Propaganda

After the initial attack, a public signal partitions B along $b_1 - b_2 = \Delta$:

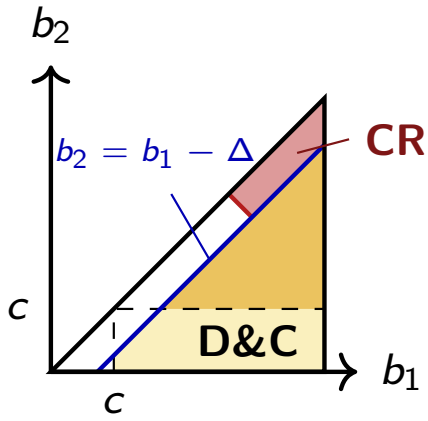
$$L_\Delta := \{(b_1, b_2) \in B : b_1 - b_2 > \Delta\} \text{ (Particularizing)}, \quad H_\Delta := B \setminus L_\Delta$$

Proposition 7(i): Propaganda ($c < \hat{c}$)

- **Without propaganda ($c < \hat{c}$)**: only CR; only $J(c) = \{p_2(b_1 + b_2) \geq 2c\}$ attack.
- **With propaganda**: $\exists\, \bar{\Delta}(c) \in (0, 1-c)$ s.t. if L_Δ is sufficiently particularizing ($\Delta \in (\bar{\Delta}(c), 1-c)$), the signal induces **D&C on L_Δ** while leaving **CR on H_Δ**.
- **Implication**: Attack set becomes $(L_\Delta \cap \{b_1 \geq c\}) \cup (H_\Delta \cap J(c)) \Rightarrow$ carves out a new D&C region below $b_2 = b_1 - \Delta$ and **expands the attack set**.



Without propaganda



With propaganda

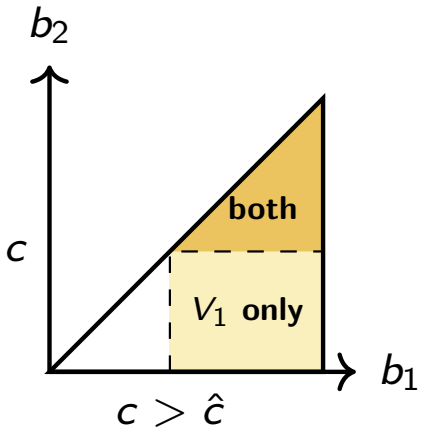
Propaganda and Counter-Propaganda

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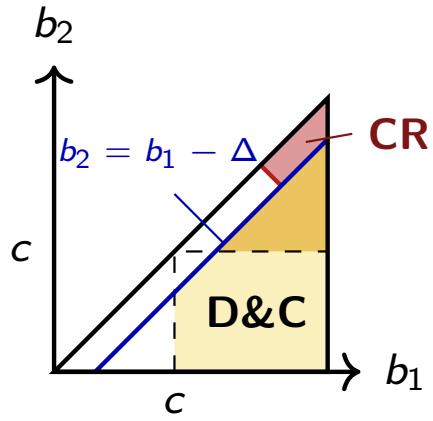
$$H_\Delta := \{(b_1, b_2) \in B : b_1 - b_2 \leq \Delta\} \text{ (Systemic)}$$

Proposition 7(ii): Counter-Propaganda ($c > \hat{c}$)

- **Without ($c > \hat{c}$)**: only attacker-preferred D&C; $\{b_1 \geq c\} \cap B$ attack.
- **With counter-propaganda**: $\exists\, \underline{\Delta}(c) > 0$ s.t. if H_Δ is sufficiently systemic ($\Delta \in (0, \underline{\Delta}(c))$), the signal induces **CR on** H_Δ while leaving **D&C on** L_Δ.
- **Implication**: Types above $b_2 = b_1 - \Delta$ now face collective resistance (only $H_\Delta \cap J(c)$ attack) $\Rightarrow$ **shrinks the attack set**.



Without counter-propaganda



With counter-propaganda

Treaty Making: Wedge Widening

Institutions alter both material incentives and inference (penalty $\gamma > 0$).

Proposition 8 (Neutrality / Non-Aggression Treaty between A and V_2)

For $c < \hat{c}$ (CR without treaty), a treaty raising A's cost of attacking V_2 to $c + \gamma$ and V_2's cost of joining to $p_2\ell + d + \gamma$ induces **D&C** whenever:

$$\Pr(b_2 \geq c + \gamma \mid b_1 \geq c)\,\ell \leq p_2\ell + d + \gamma.$$

- **Perverse effect**: Reassures V_2 and exposes V_1 (attack set expands from $J(c)$ to $\{b_1 \geq c\}$). For small p_2, it can even **increase the probability that V_2 herself is eventually attacked**!

Treaty Making: Wedge Closing

Institutions alter both material incentives and inference (penalty $\gamma > 0$).

Proposition 9 (Collective-Defense Treaty between V_1 and V_2)

For $c > \hat{c}$ (D&C without treaty), a treaty imposing penalty γ on refusing to join V_1 eliminates D&C and restores **CR** whenever:

$$\gamma > \ell\big[\kappa - \phi(c)\big].$$

- Does not need to raise $\phi(c)$; makes inaction costly even when $\phi(c)$ is small.

More Than 2 Victims

What if V_2 can rely on a downstream victim V_3? $(b_1 \geq b_2 \geq b_3,\ p_2 \leq .5)$.

- If V_2 is attacked later, a **fallback alliance** with V_3 is available w.p. $t \in [0,1]$.
- A attacks V_2 later iff $(b_1, b_2, b_3) \in K_t(c)$:

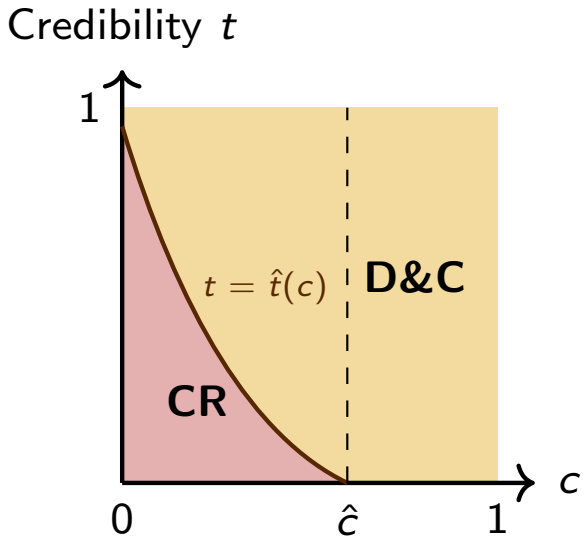
$$(1-t)[b_2 - c + (b_3 - c)_+] + t[p_2(b_2 + b_3) - 2c] \geq 0.$$

Proposition 10

For $c < \hat{c}$, there is a unique $\hat{t}(c) \in (0,1)$ s.t.

$$\left[1 - (1-\kappa)\hat{t}(c)\right] \Pr\left(K_{\hat{t}(c)}(c) \mid b_1 \geq c\right) = \kappa.$$

For $t > \hat{t}(c)$, **downstream protection switches CR to D&C**, isolating V_1!



Insight: Protected neighbors feel safer waiting behind their own defense line, which can expose an unprotected V_1.

Conclusion

- **Core Mechanism: The Bayesian Wedge**
  - Victims divide not from mechanical coordination failure, but from **uncertainty about target scope**.
  - Generates D&C and uncertain collective resistance.
- **What Facilitates or Prevents Division?**
  - **Inference**: Higher attack cost makes an observed strike more exceptional, triggering an **upward jump in aggression** at $\hat{c}$ (reinforced by lower correlation ρ, higher specialness a, and particularizing propaganda).
  - **Psychological Distortions**: Wishful thinking and magical thinking expand D&C beyond the Bayesian benchmark.
  - **Institutions**: Collective-defense treaties restore CR by making inaction costly; neutrality pacts and downstream protection reassure the bystander and isolate V_1.
- Effective collective defense requires both the ability to fight together **and** a common interpretation of what the first attack reveals.

Thank You!

Questions or comments?

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Paper available at arXiv:2607.12371 | https://jinyuqi-huang.github.io/

Related Literature

- Divide-and-Conquer & Discriminatory Multi-Agent Contracting:
  - Segal (2003); Winter (2004); Halac, Kremer, and Winter (2020).
  - Those models use discriminatory offers to overcome coordination.
- Rationalist Explanations for War, Appeasement & Deterrence:
  - Fearon (1995); Powell (2006); Gurantz and Hirsch (2017); Ripsman and Levy (2008); Maass (2021).
  - *Contribution*: Formalizes how target-specific vs. systemic uncertainty endogenously generates sequential conquest and uncertain collective resistance.
- Motivated Beliefs:
  - Brunnermeier and Parker (2005).
- Strategic Disclosure:
  - Grossman (1981); Milgrom (1981); Milgrom and Roberts (1986).

◂ Back to Overview

Backup: Mixed Equilibrium Characterization (OA.4)

Let $\alpha = \Pr(V_1 \text{ fights})$ and $\beta = \Pr(V_2 \text{ joins})$.

- V_1 is indifferent between fighting and surrendering when:

$$\beta = \beta_0 := \frac{p_1 + d/\ell - 1}{p_1 - p_2} \in (0, 1).$$

- Given (α, β), A's expected payoff from attacking V_1 is:

$$(1-\alpha)\{b_1 - c + (b_2 - c)_+\} + \alpha(1-\beta)\{p_1 b_1 - c + (b_2 - c)_+\} + \alpha\beta\{p_2(b_1 + b_2) - 2c\}.$$

- Define $B_{\alpha,\beta}(c)$ as the set of types where this payoff is ≥ 0. A mixed equilibrium satisfies:

$$0 < \alpha \leq 1, \quad \beta_0 \leq \beta < 1, \quad (\alpha < 1 \implies \beta = \beta_0),$$

$$\text{and} \quad \Pr(b_2 \geq c \mid B_{\alpha,\beta}(c)) = \kappa.$$

- **Conditional Outcomes**: Given an initial attack, joint resistance occurs w.p. $\alpha\beta$; an attack confined to V_1 occurs w.p. $(1 - \alpha\beta)(1 - \kappa)$; sequential attack on V_2 occurs w.p. $(1 - \alpha\beta)\kappa$.

◂ Back